CSE 331: Computer Security Fundamentals

Spring 2026

Basic Cryptography:
Randomness, Historical Ciphers, Alice and Bob Settings

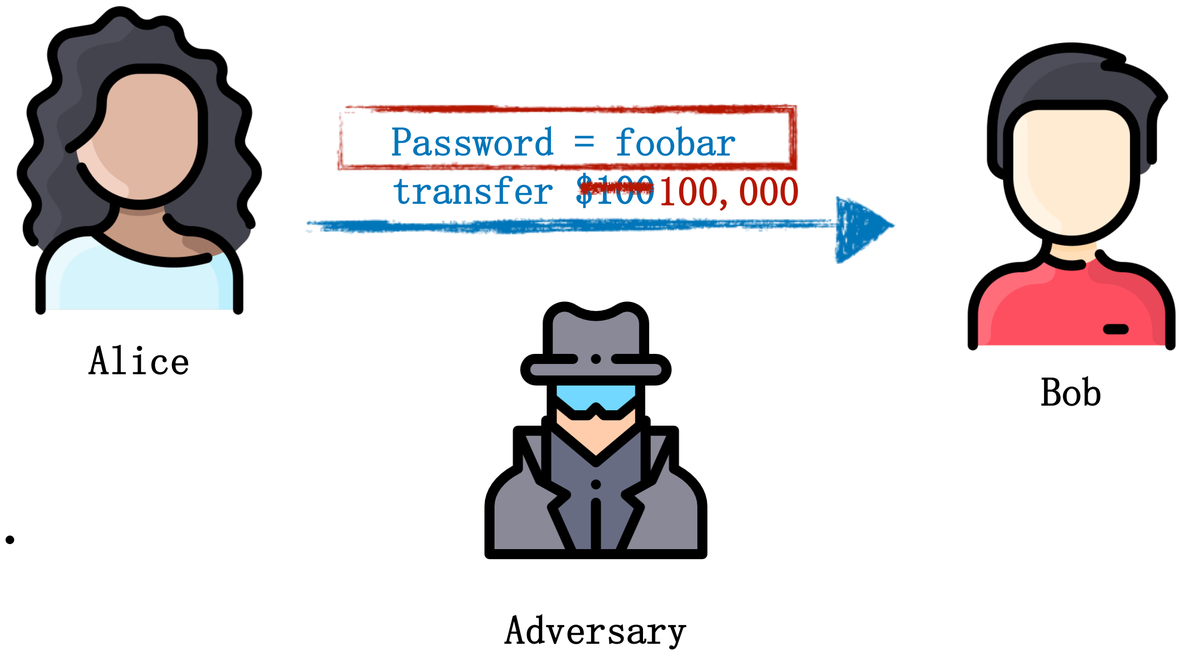
Amir Rahmati
Xinrui Yu

Stony Brook Institute
at Anhui University
安徽大学纽约石溪学院

Communication Security Goals

- **Confidentiality** of data:
  Prevent exposure of information.

- **Integrity** of data:
  Prevent modification of information.



Recall the Bigger Picture

- Cryptography is only one piece of the puzzle.
  - Physical security
  - OS security
  - Network security
  - Users
  - Cryptography
- A chain is as strong as its **weakest link.**
- Still, cryptography is a crucial tool in our toolbox.

Security Through Obscurity

- Reliance on design or implementation secrecy as the main method of providing security to a system or component.

"Rogues are very keen in their profession, and know already much more than we can teach them."
— Alfred Charles Hobbs,
"Locks & Keys" 1853

"Security Through Obscurity" is an **illusion**.

Kerckhoff's Principle

- Security of a cryptographic object should **depend only on the secrecy of the secret (private) key.**

- Security should not depend on the secrecy of the algorithm itself.

- Shannon's Maxim:
  "the enemy knows the system"

  "One ought to design systems under the assumption that the enemy will immediately gain full familiarity with them."

"the system must not require secrecy and must be able to be stolen by the enemy without causing trouble."
— Auguste Kerckhoffs,
"La Cryptographie Militaire", 1883

Randomness

```
int getRandomNumber()
{
    return 4;  // chosen by fair dice roll.
               // guaranteed to be random.
}
```

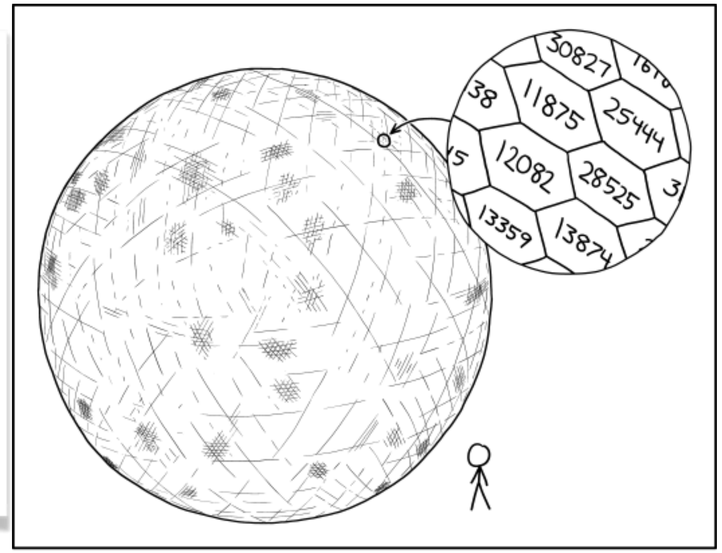
- Many applications (especially security ones) require randomness.
  - Generate secret cryptographic keys
  - Generate random initialization vectors for encryption
  - Generate passwords for new users
  - Determine results of random events (critical hits, gacha games, etc.)

How are random numbers generated?

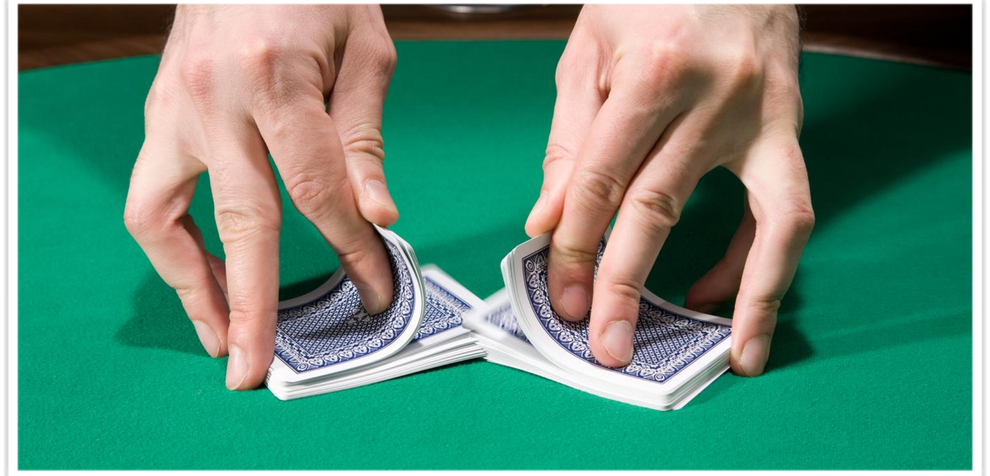
- Ideally:
  - Uniform distribution
  - Statistically independent

- True Random Number Generators (TRNGs)
  - Extract randomness from physical phenomena
- Pseudo-Random Number Generators (PRNGs)
  - Algorithms that use mathematical formulae to produce sequences of numbers that appear random.
  - Efficient, deterministic, periodic

Sources of Entropy



THE HARDEST PART OF SECURELY GENERATING RANDOM 16-BIT NUMBERS IS ROLLING THE D65536.



RANDOM.ORG

LAVARND™
...truly random since 2000

HotBits: **Genuine random numbers, generated by radioactive decay**

Famous Example: Cloudflare and its lava lamps

C's rand() Function

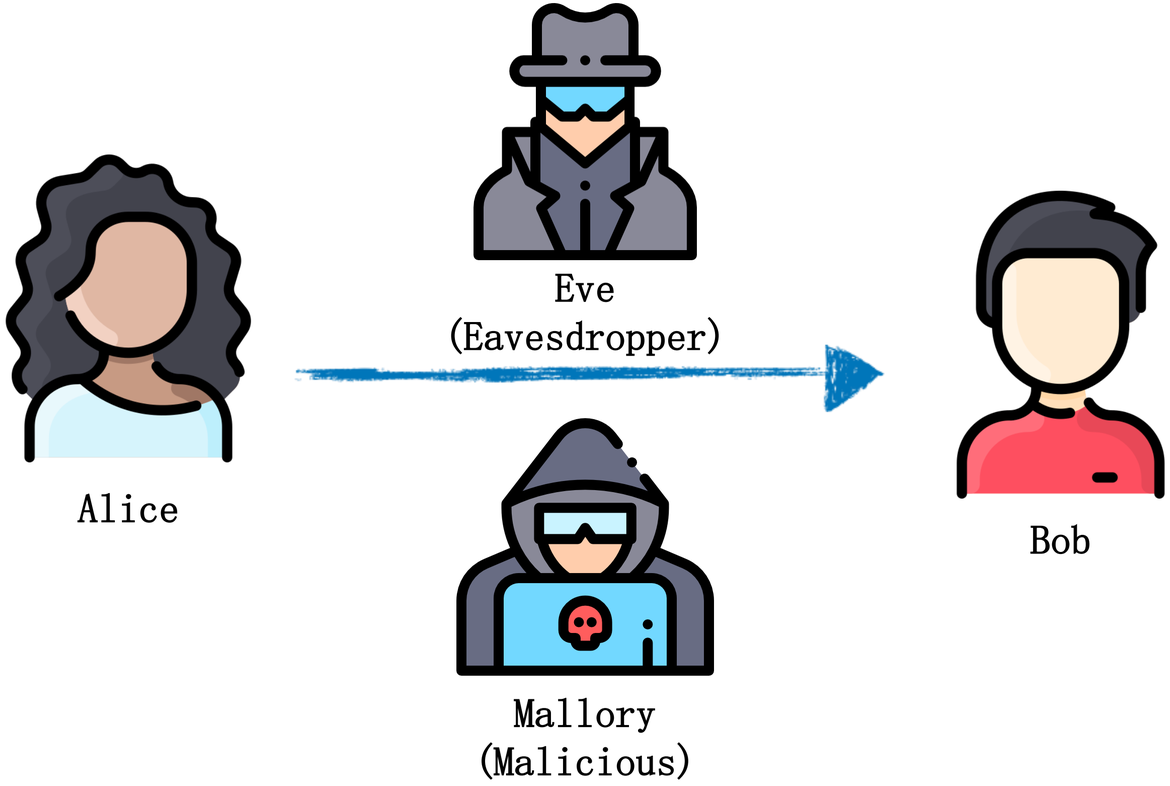
- C has a built-in random function: rand()

```
unsigned long int next = 1;
/* rand:  return pseudo-random integer on 0..32767 */
int rand(void) {
    next = next * 1103515245 + 12345;
    return (unsigned int)(next/65536) % 32768;
}
/* srand:  set seed for rand() */
void srand(unsigned int seed) {
    next = seed;
}
```

- Problem: Given a few sample outputs, you can predict subsequent ones

Alice and Bob

and Mallory, Eve, Trent, Carole, Dave, Trudy, Olivia, Victor…



I'VE DISCOVERED A WAY TO GET COMPUTER SCIENTISTS TO LISTEN TO ANY BORING STORY.

The Origin

"... For our scenarios we suppose that A and B (also known as Alice and Bob) are two users of a public-key cryptosystem. We will distinguish their encryption and decryption procedures with subscripts: E_A, D_A, E_B, D_B."

Received April 4, 1977

A Method for Obtaining Digital Signatures and Public-Key Cryptosystems

R.L. Rivest, A. Shamir, and L. Adleman*

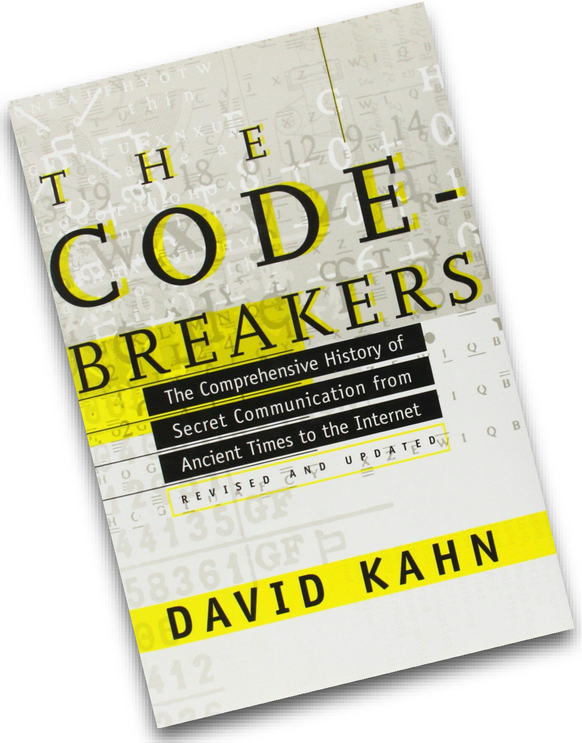
Abstract

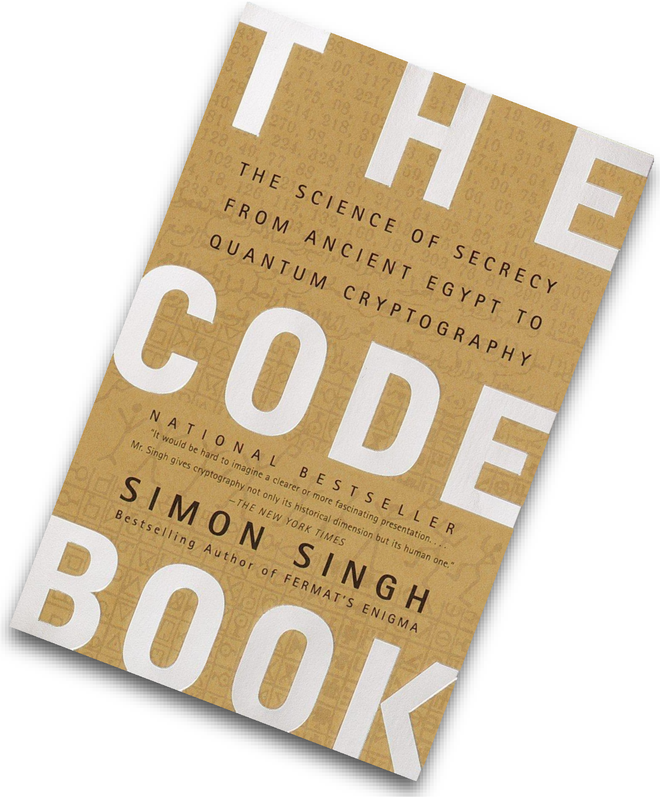
An encryption method is presented with the novel property that publicly revealing an encryption key does not thereby reveal the corresponding decryption key. This has two important consequences:

1. Couriers or other secure means are not needed to transmit keys, since a message can be enciphered using an encryption key publicly revealed by the intended recipient. Only he can decipher the message, since only he knows the corresponding decryption key.
2. A message can be "signed" using a privately held decryption key. Anyone can verify this signature using the corresponding publicly revealed encryption key. Signatures cannot be forged, and a signer cannot later deny the validity of his signature. This has obvious applications in "electronic mail" and "electronic funds transfer" systems.

Classic Cryptos

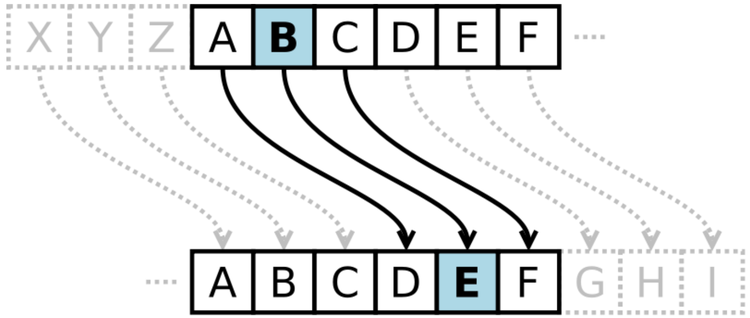
- Substitution Ciphers
- Transposition Ciphers
- Codebooks
- Machines





Caesar Cipher (Shift Cipher)

- Plaintext letters are replaced with letters a fixed shift away in the alphabet.



- Example:
  - Plaintext: `The quick brown fox jumps over the lazy dog`
  - Key: Shift 3

    `ABCDEFGHIJKLMNOPQRSTUVWXYZ`
    `DEFGHIJKLMNOPQRSTUVWXYZABC`
  - Ciphertext: `WKHTX LFNEU RZQIR AMXPS VRYHU WKHOD CBGRJ`

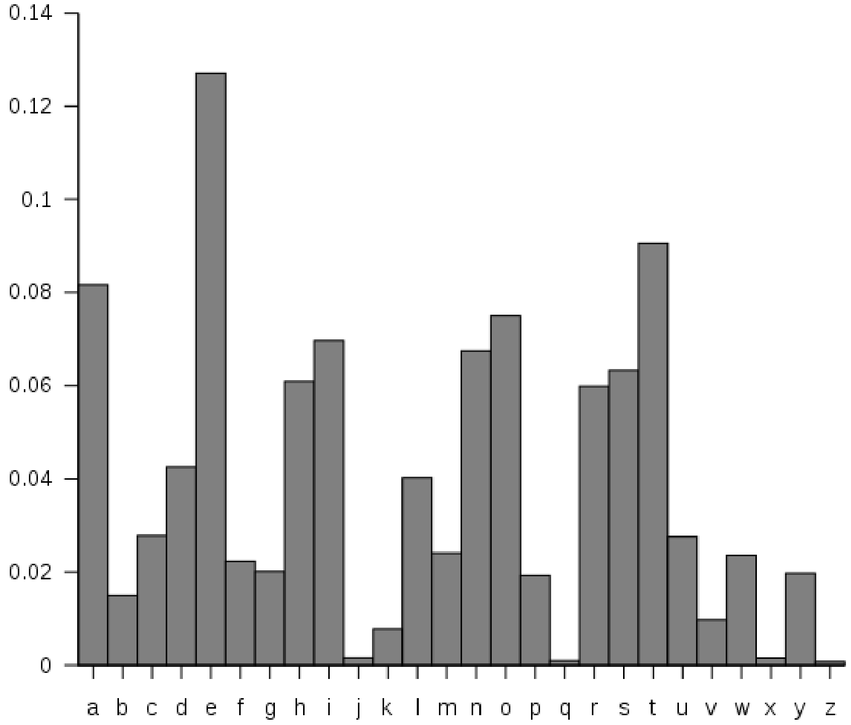
Simple Substitution Cipher

- Superset of shift ciphers: each letter is substituted for another one.

- Add a secret key
  - Plaintext Alphabet:
    `ABCDEFGHIJKLMNOPQRSTUVWXYZ`
  - Cipher Alphabet:
    `ZEBRASCDFGHIJKLMNOPQTUVWXY`

Simple Substitution Cipher

- What is the key space? 26! ~= 2^88
- How to attack?
  - Frequency analysis.



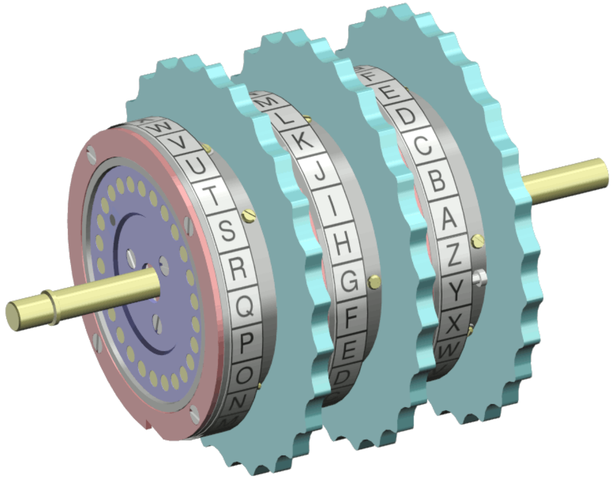
Bigrams:

th 1.52%	en 0.55%	ng 0.18%
he 1.28%	ed 0.53%	of 0.16%
in 0.94%	to 0.52%	al 0.09%
er 0.94%	it 0.50%	de 0.09%
an 0.82%	ou 0.50%	se 0.08%
re 0.68%	ea 0.47%	le 0.08%
nd 0.63%	hi 0.46%	sa 0.06%
at 0.59%	is 0.46%	si 0.05%
on 0.57%	or 0.43%	ar 0.04%
nt 0.56%	ti 0.34%	ve 0.04%
ha 0.56%	as 0.33%	ra 0.04%
es 0.56%	te 0.27%	ld 0.02%
st 0.55%	et 0.19%	ur 0.02%

Trigrams:

1. the	6. ion	11. nce
2. and	7. tio	12. edt
3. tha	8. for	13. tis
4. ent	9. nde	14. oft
5. ing	10.has	15. sth

Enigma Machine



Uses rotors (substitution cipher) that change position after each key.

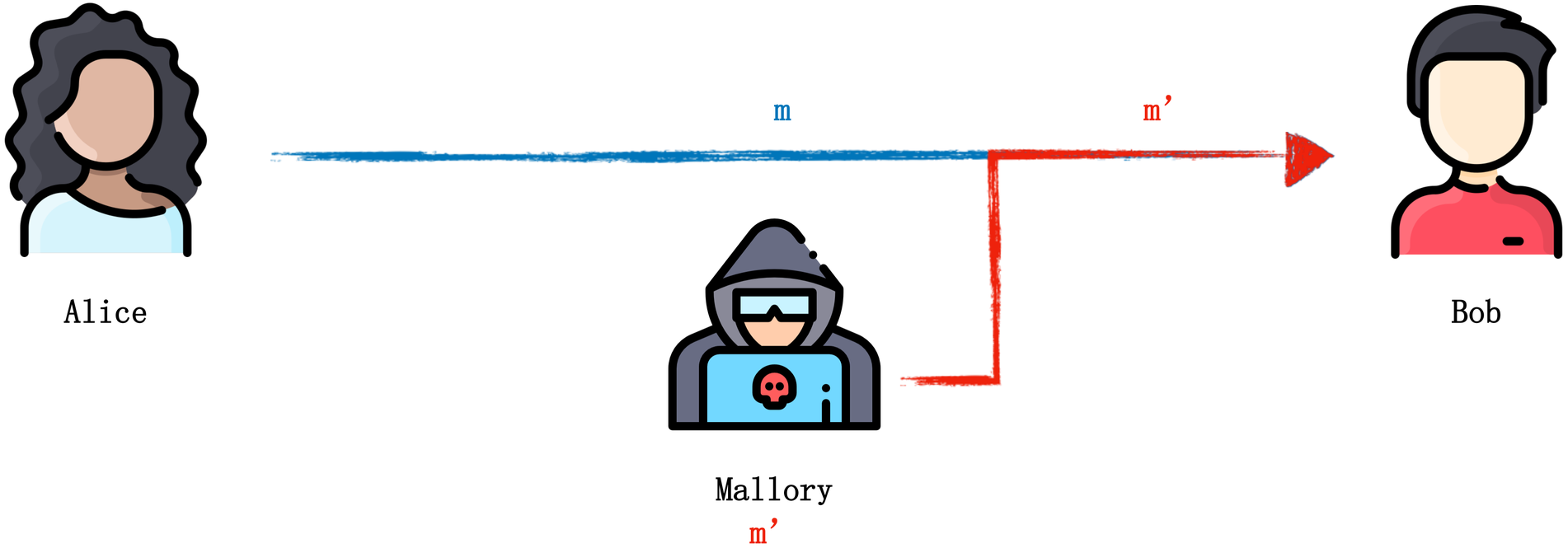
2xn+1 substitutions

Use plugs to do an additional round of substitution.

0/1/2 substitutions

Key space?

60 rotor combinations x 26^n for n rotors x 151 trillion wire combinations

Attack on Confidentiality

Attack on Integrity